

SIEM & XDR Implementation with Wazuh - Lab Report

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Course: Cyber Defense

Estimated Duration: 8–10 hours

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Executive Summary

In this comprehensive lab exercise, we implemented and compared both local and cloud-based SIEM deployment methods using Wazuh. The local deployment (Lab 1) provided complete control over infrastructure and data but required manual resource management and ongoing maintenance. In contrast, the cloud deployment (Lab 2) offered faster setup times, automatic scalability, and eliminated the need for infrastructure management, making it ideal for organizations seeking a managed security solution without capital expenses.

Lab 1: Local Hypervisor Deployment (On-Premise SIEM)

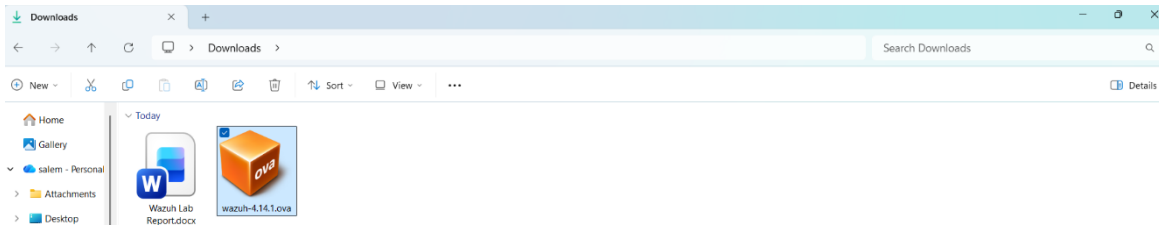
Objective

To deploy a fully functional Wazuh stack (Indexer, Server, and Dashboard) using a pre-built virtual appliance on a local hypervisor (Oracle VirtualBox) and perform baseline security monitoring with agent deployment and alert generation.

Section 1: Environment Setup

Step 1: Download Wazuh OVA File

1. Navigate to: <https://packages.wazuh.com/4.x/vm/wazuh-4.14.1.ova>
2. Download the Wazuh OVA file (approximate size: 3-4 GB)
3. After download completion, verify file integrity using SHA-256 checksum provided on the Wazuh packages page
4. Store the OVA file in an easily accessible location (e.g., Documents folder or dedicated VM folder)



Downloaded Wazuh OVA File

Step 2: Import OVA into VirtualBox

1. Open Oracle VM VirtualBox (ensure you have version 7.0 or later)
2. From the **File** menu, select **Import Appliance**
3. Click the folder icon and browse to select the downloaded wazuh-4.14.1.ova file
4. Click **Next** to proceed to the appliance settings page
5. Review and verify the hardware settings:
 - **RAM:** Minimum 4096 MB (Recommended: 8192 MB for better performance)
 - **CPU:** Minimum 2 vCPUs (Recommended: 4 vCPUs)
 - **Disk Space:** 50 GB (50 GB VMDK file)
 - **Network:** Leave as default for now (will configure in next step)
6. Click **Import** and wait for the import process to complete (this may take 5-10 minutes depending on disk speed)
7. Monitor the import progress bar - do not interrupt the process

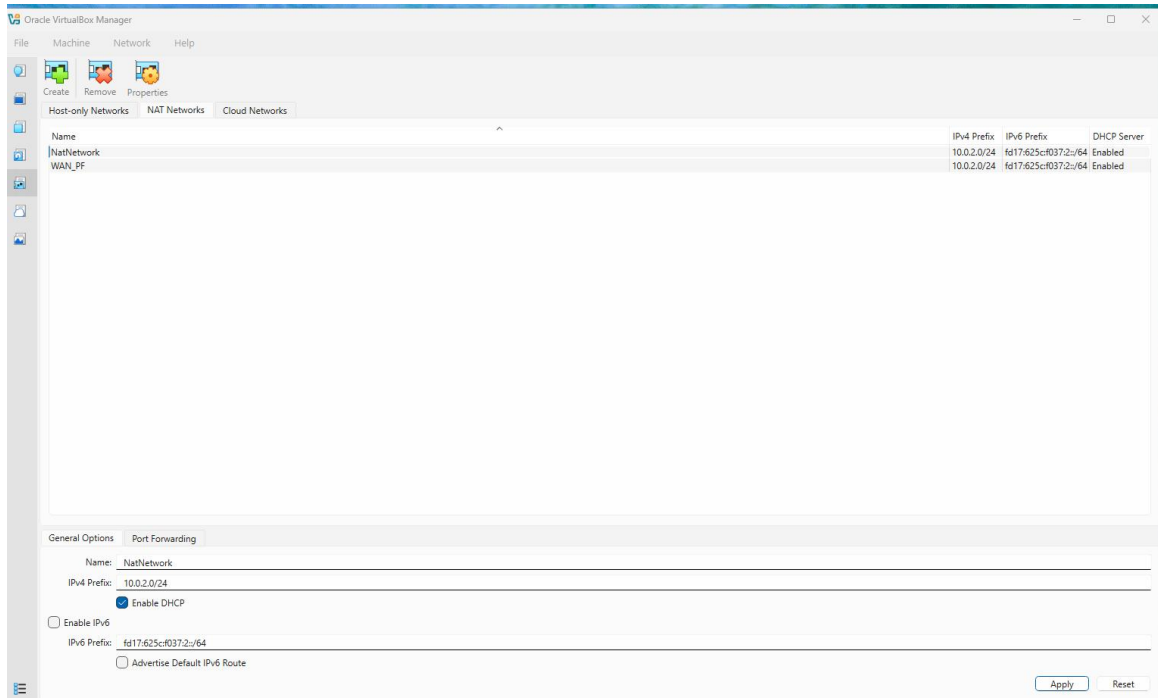
Step 3: Configure Network Adapter to NAT Network Mode

Important: Before starting the VM, you must create and configure a NAT Network in VirtualBox.

Creating NAT Network:

1. In VirtualBox, go to **File > Tools > Network Manager**
2. In the Network Manager window, select the **NAT Networks** tab
3. Click the **Create** button to create a new NAT Network
4. A new NAT Network will be created with default settings:

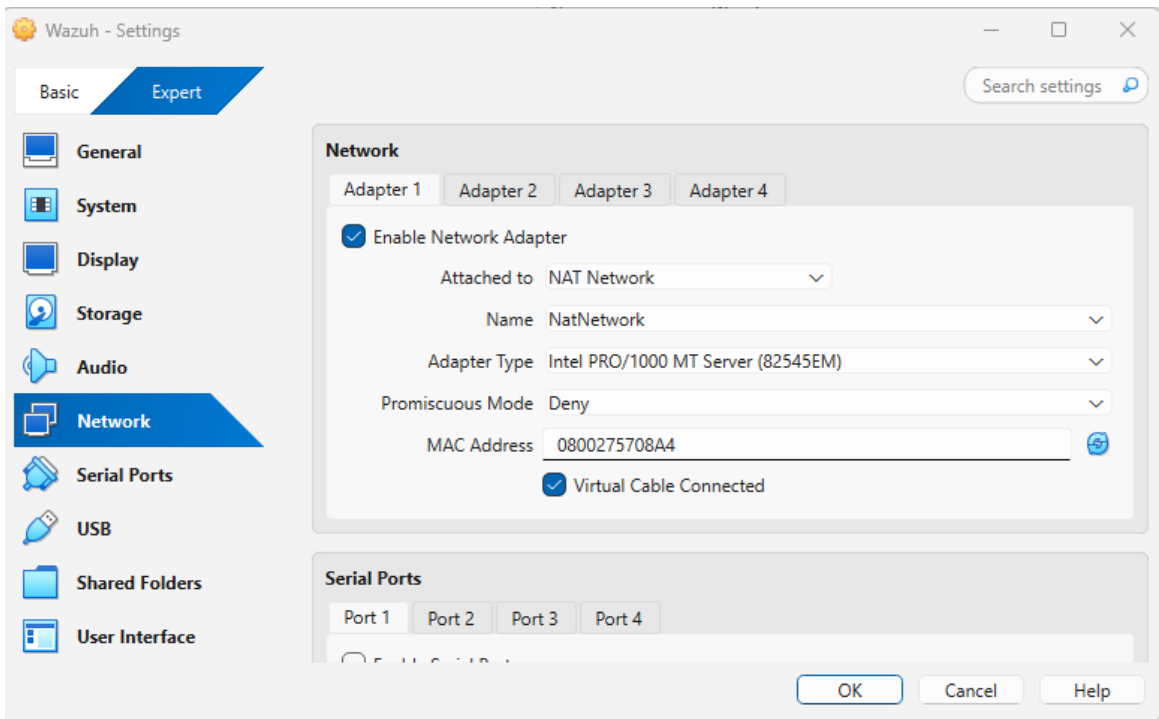
- Name: "NatNetwork"
 - CIDR: 10.0.2.0/24
 - DHCP: Enabled
5. Verify that **Supports DHCP** checkbox is checked (enabled)
 6. Click **Apply** and then **OK** to save the settings



VirtualBox NAT Network Configuration

Configuring Wazuh VM Network:

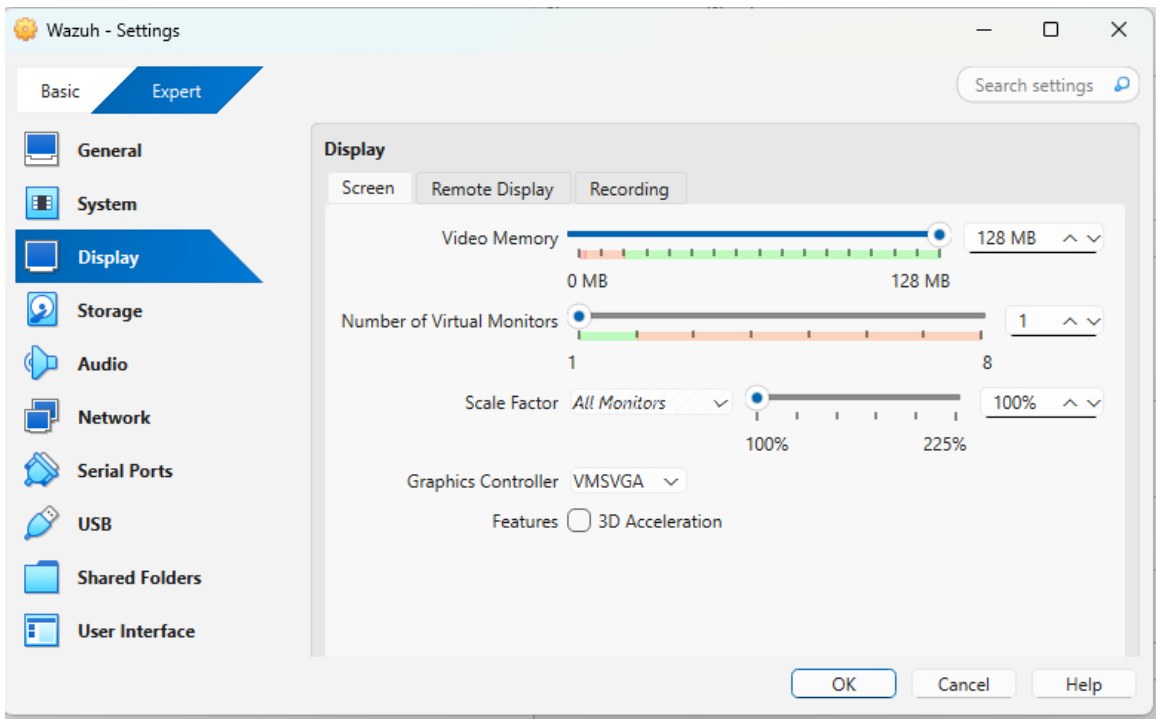
1. Right-click on the Wazuh VM and select **Settings** (or select the VM and click Settings button)
2. Navigate to the **Network** tab
3. Select **Adapter 1** (should already be selected)
4. In the **Attached to:** dropdown menu, select **NAT Network**
5. In the **Name:** dropdown menu, select **NatNetwork** (the one we just created)
6. Click **OK** to apply the network settings
7. The Wazuh VM is now configured to use the NAT Network



VirtualBox Network Settings - NAT Network Mode

Step 4: Configure Graphics Controller (Important for VirtualBox)

1. Select the Wazuh VM (do not start it yet)
2. Click the **Settings** button
3. Navigate to the **Display** tab
4. At the top of the window, change from **Basic** to **Expert** mode to access all settings
5. In the **Graphics Controller** dropdown, select **VMSVGA** (recommended for Linux VMs in VirtualBox)
6. Set **Video Memory** to at least 128 MB (or higher if available)
7. Leave other display settings as default
8. Click **OK** to save the graphics settings



VirtualBox Display Settings - VMSVGA

Section 2: Initialization and VM Access

Step 5: Start the Wazuh Virtual Machine

1. Select the Wazuh VM in VirtualBox
2. Click the **Start** button (green arrow) to power on the VM
3. The VM will boot Linux and display the login prompt (this typically takes 2-3 minutes)
4. Wait for the system to complete booting - you will see:
 - BIOS/UEFI startup messages
 - Linux kernel loading messages
 - Login prompt with "wazuh-user login:"
5. At the login prompt, enter the credentials:

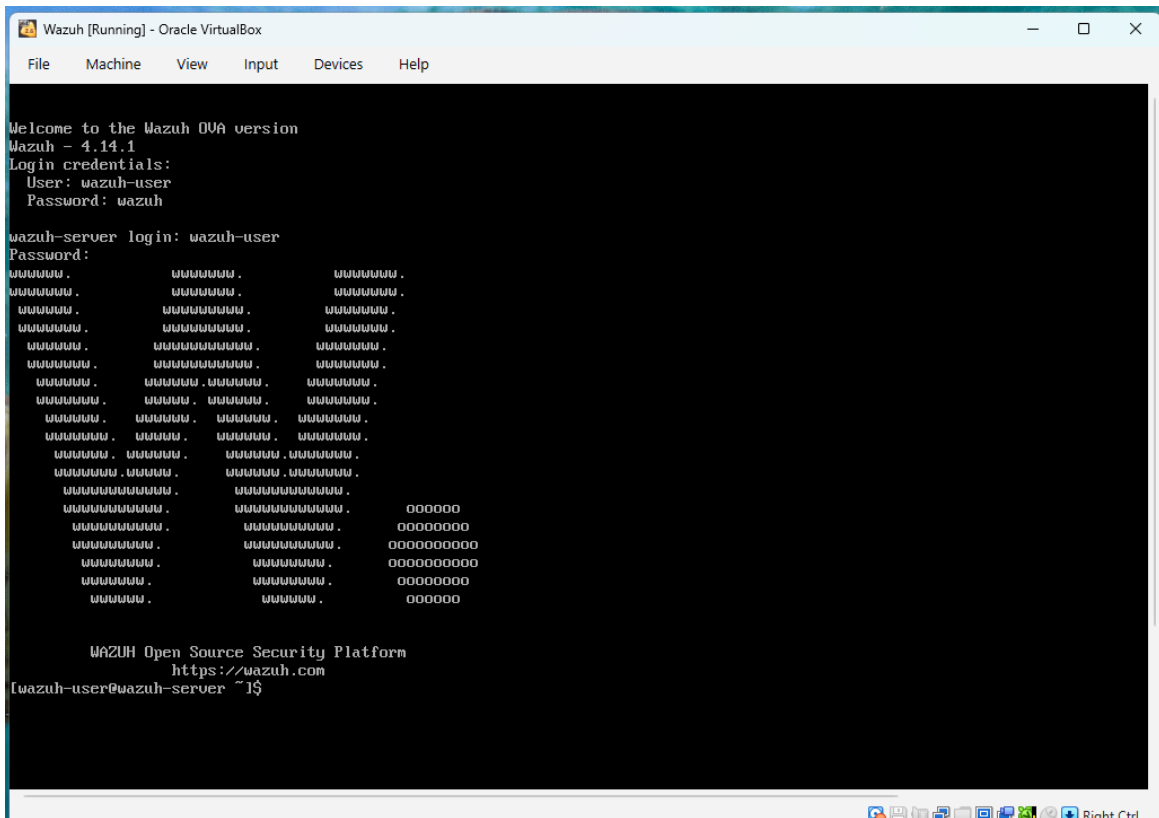
Username: wazuh-user

Password: wazuh

6. After successful login, you will see the Linux command prompt
7. Type the following command to display the network configuration and IP address:

ip a

8. Locate the IP address assigned to the eth0 interface (typically in the 10.0.2.x range, e.g., 10.0.2.15)
9. Write down or take note of this IP address - you will need it for the next steps



Wazuh VM Console

```
wazuh-user@wazuh-server ~$ ip a
: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:57:08:a4 brd ff:ff:ff:ff:ff:ff
    altname enp0s17
    inet 10.0.2.3/24 metric 1024 brd 10.0.2.255 scope global dynamic eth0
        valid_lft 453sec preferred_lft 453sec
    inet6 fe80::a00:27ff:fe57:8a4/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
wazuh-user@wazuh-server ~$
```

IP Address Display

Step 6: Access Wazuh Dashboard via Web Browser

1. On your host machine, open a web browser (Chrome, Firefox, Edge, Safari, etc.)
2. In the address bar, enter the URL using the IP address obtained from Step 5:

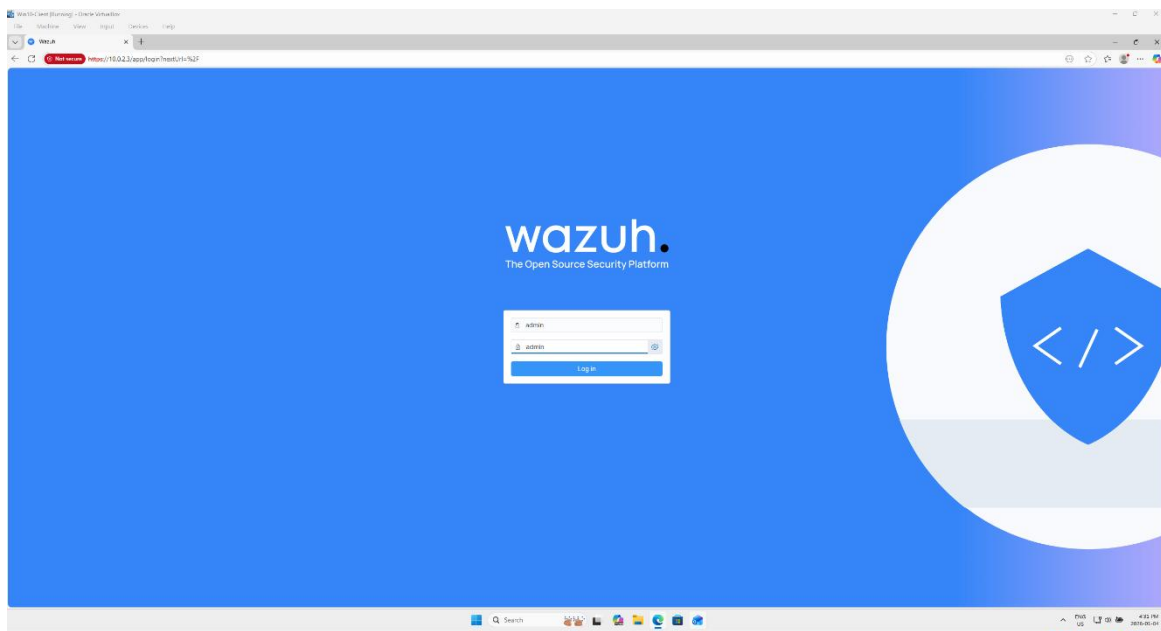
<https://10.0.2.3>

3. The browser will display an SSL/TLS certificate warning because the Wazuh server uses a self-signed certificate
4. Click **Advanced** button on the warning page
5. Click **Proceed to 10.0.2.3** (or "Accept the Risk and Continue" in Firefox)
6. The Wazuh Dashboard login page should now appear
7. At the login page, enter the default credentials:

Username: admin

Password: admin

8. Click **Log In** or press Enter
9. You should now be logged into the Wazuh Dashboard



Wazuh Dashboard Login Page

Section 3: Security Baseline Configuration

Step 7: Change Admin Password (Mandatory for Security Compliance)

1. Return to the Wazuh VM console window
2. Press **Ctrl+C** or type **exit** if you need to log out (then log back in with original credentials)
3. Execute the password change command:

```
sudo /usr/share/wazuh-indexer/plugins/opensearch-security/tools/wazuh-passwords-tool.sh -u admin -p Pa$$w0rd*
```

(Replace "NewSecurePassword123!" with your chosen secure password)

4. The command will execute and update the admin password in the Wazuh system
5. Return to the Wazuh Dashboard in your browser
6. Click **Log Out** or refresh the page
7. Log in again with the new password to verify the change was successful

Note: This password change is mandatory for security compliance and should be done immediately after initial setup.

```
wazuh-user@wazuh-server:~$ sudo /usr/share/wazuh-indexer/plugins/opensearch-security/tools/wazuh-passwords-tool.sh -u admin -p Pa$$w0rd*
05/01/2026 20:03:17 INFO: Updating the internal users.
05/01/2026 20:03:21 INFO: A backup of the internal users has been saved in the /etc/wazuh-indexer/internalusers-backup folder.
05/01/2026 20:03:21 INFO: Generating password hash
05/01/2026 20:03:26 INFO: The filebeat.yml file has been updated to use the Filebeat Keystore username and password.
05/01/2026 20:03:50 WARNING: Password changed. Remember to update the password in the Wazuh dashboard, Wazuh server, and Filebeat nodes if necessary, and restart the services.
```

Change Admin Password

Step 8: Enable Vulnerability Detection and File Integrity Monitoring (FIM) Modules

Enabling Vulnerability Detection:

1. In the Wazuh VM console, open the configuration file:

```
sudo nano /var/ossec/etc/ossec.conf
```

2. Use Ctrl+W to search for "vulnerability-detection"
3. Locate the vulnerability-detection section and verify it looks like this:

```
yes yes 60m
```

4. If "enabled" is set to "no", change it to "yes"
5. Save the file by pressing Ctrl+X, then Y, then Enter
6. Restart the Wazuh Manager to apply changes:

`sudo systemctl restart wazuh-manager`

7. Verify the service restarted successfully:

`sudo systemctl status wazuh-manager`

```
<vulnerability-detection>
  <enabled>yes</enabled>
  <index-status>yes</index-status>
  <feed-update-interval>60m</feed-update-interval>
</vulnerability-detection>
```

Vulnerability Detection

Verify File Integrity Monitoring:

1. FIM is enabled by default in Wazuh
2. To verify FIM settings, in the Wazuh VM console, view the configuration:

`sudo nano /var/ossec/etc/ossec.conf`

3. Search for the "syscheck" section which contains FIM configuration
4. FIM typically monitors critical system directories like /etc, /usr/bin, /usr/sbin, etc.
5. Exit the file (Ctrl+X without saving changes)

```
<!-- Directories to check (perform all possible verifications) -->
<directories>/etc,/usr/bin,/usr/sbin</directories>
<directories>/bin,/sbin,/boot</directories>
```

File Integrity Monitoring (FIM)

Section 4: Agent Deployment

Step 9: Deploy Wazuh Agent on Windows Host Machine

1. In the Wazuh Dashboard (in your browser), click the menu icon (three horizontal lines) on the left sidebar
2. Navigate to **Agents > Agent management** or simply select **Agents**
3. Click the **Deploy new agent** button (usually a blue button on the page)
4. A dialog/wizard will appear asking for agent configuration. Select the following:
 - **Operating System:** Windows
 - **Architecture:** 64-bit (x86_64) [or 32-bit if your Windows system is 32-bit]
 - **Server Address:** Enter the IP address from Step 5 (e.g., 10.0.2.3)

- **Agent Name:** Windows-Host-Agent (or any descriptive name)
5. The wizard will generate a PowerShell installation command. Copy this entire command to your clipboard

generated command:

```
Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.14.1-1.msi -OutFile $env:tmp\wazuh-agent; msixexec.exe /i $env:tmp\wazuh-agent /q WAZUH_MANAGER='10.0.2.3' WAZUH_AGENT_NAME='Windows-Host-Agent'
```

On your Windows host machine, open PowerShell with Administrator privileges:

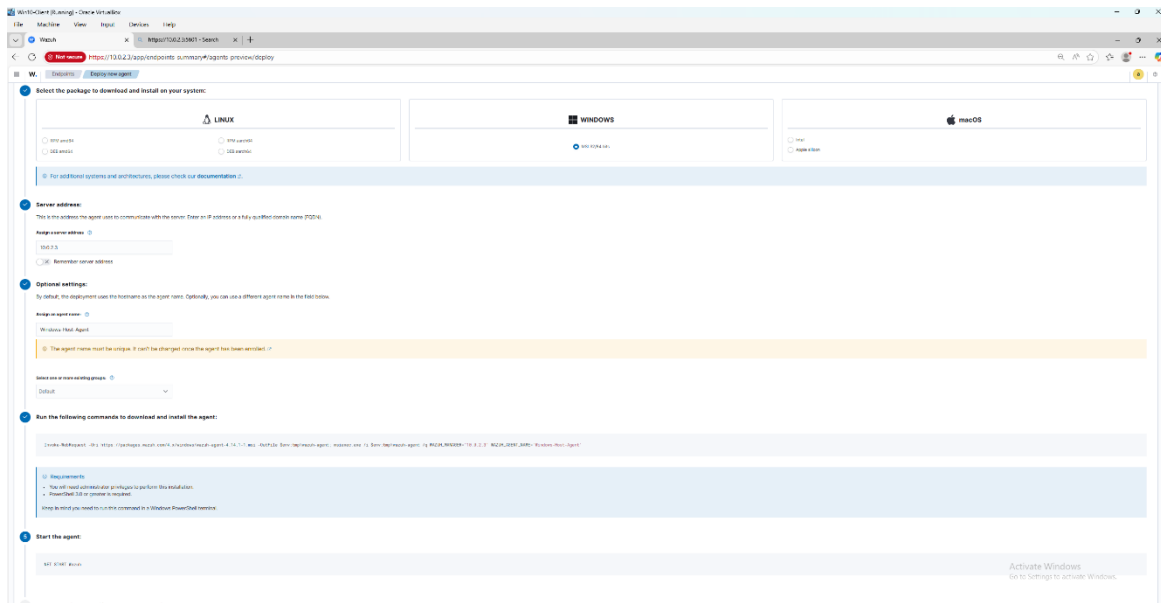
- Press **Windows key + R**
 - Type: powershell
 - Press **Ctrl+Shift+Enter** to open as Administrator
 - Click **Yes** if prompted by User Account Control
7. Paste the copied command into PowerShell and press **Enter**
 8. The installation will proceed automatically. Wait for it to complete (typically 1-2 minutes)
 9. Once installation finishes, start the Wazuh Agent service:

```
NET START WazuhSvc
```

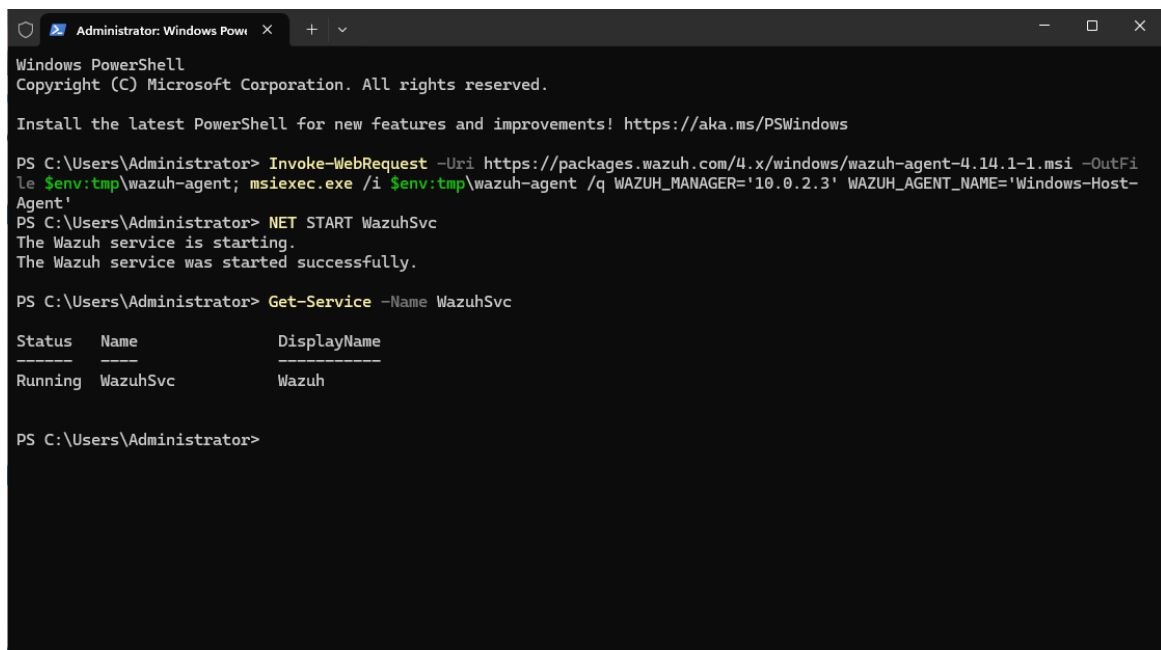
10. Verify the service started successfully:

```
Get-Service -Name WazuhSvc
```

Status should show as "Running"



Wazuh Deploy New Agent Wizard



PowerShell Installation Process

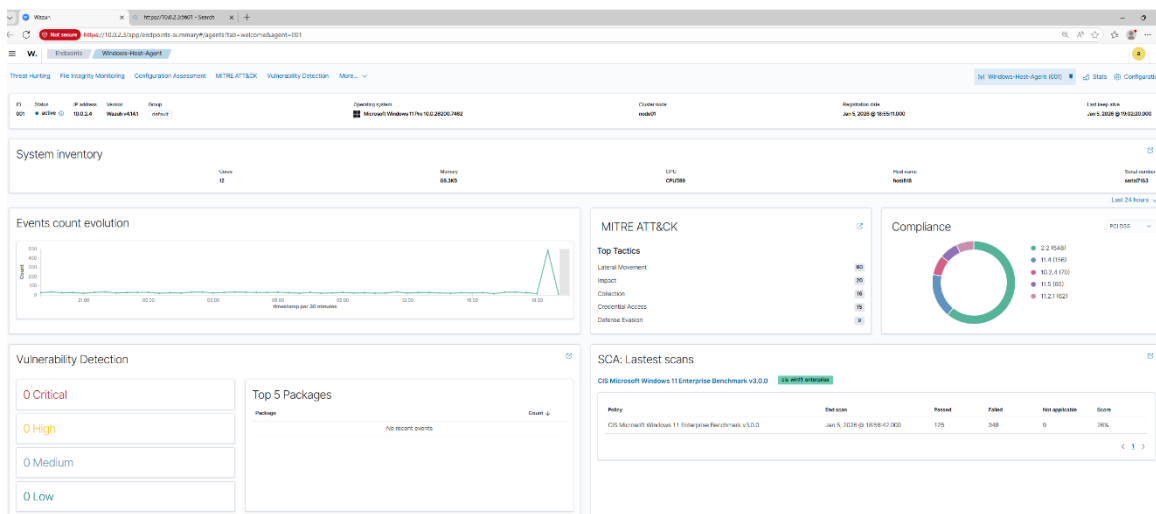
Step 10: Verify Agent is Active in Dashboard

1. Return to the Wazuh Dashboard in your browser
2. If not already there, navigate to **Agents** section
3. Look for your newly deployed agent "Windows-Host-Agent" in the agents list
4. Verify the agent status shows **Active** (typically shown with a green indicator)

5. Click on the agent name to view detailed information:

- Last keep-alive timestamp
- IP address
- Agent version
- Operating system details

6. If the agent status is not Active, wait 30-60 seconds and refresh the page. It may take time for the agent to connect



Wazuh Dashboard - Active Agent Confirmation

Section 5: Triggering and Monitoring Alerts

Step 11: Generate Security Alerts for Testing

Method 1: File Modification Alert (Recommended - Simplest Method)

On your Windows host machine where the Wazuh Agent is installed:

1. Open File Explorer and navigate to:
C:\Program Files (x86)\ossec-agent\
 2. Create a new text file in this directory:
 - Right-click in the empty space
 - Select **New > Text Document**
 - Name it: test-alert.txt
 3. Open the file and add some content:
 - Double-click the file to open it

- Type: "This is a test file for Wazuh FIM monitoring"
 - Save the file (Ctrl+S)
4. Wait 30-60 seconds for the Wazuh agent to detect the file creation
 5. Now modify the file:
 - Open it again
 - Change the content or add a line
 - Save the file
 6. This file modification should trigger a File Integrity Monitoring (FIM) alert
 7. Alternatively, you can delete the file after creating it, which will also generate an FIM alert

FIM: Recent events

Time	Path	Action	Rule description	Rule level	Rule id
Jan 5, 2025 @ 17:20:16.191	/var/ossec/queue/fim/fim.db	added	File added to the system.	5	554
Jan 5, 2025 @ 17:20:16.533	/var/ossec/queue/fim/fim.db	deleted	File deleted.	7	553
Jan 5, 2025 @ 18:50:05.893	/var/ossec/queue/fim/fim.db	modified	Integrity checksum changed.	7	550
Jan 5, 2025 @ 19:00:05.840	/var/ossec/queue/fim/fim.db	deleted	File deleted.	7	553
Jan 5, 2025 @ 19:04:52.436	/var/ossec/queue/fim/fim.db	added	File added to the system.	5	554

File Modification Alert

Method 2: Failed Login Attempt Alert (Alternative Method)

If you want to generate a different type of alert:

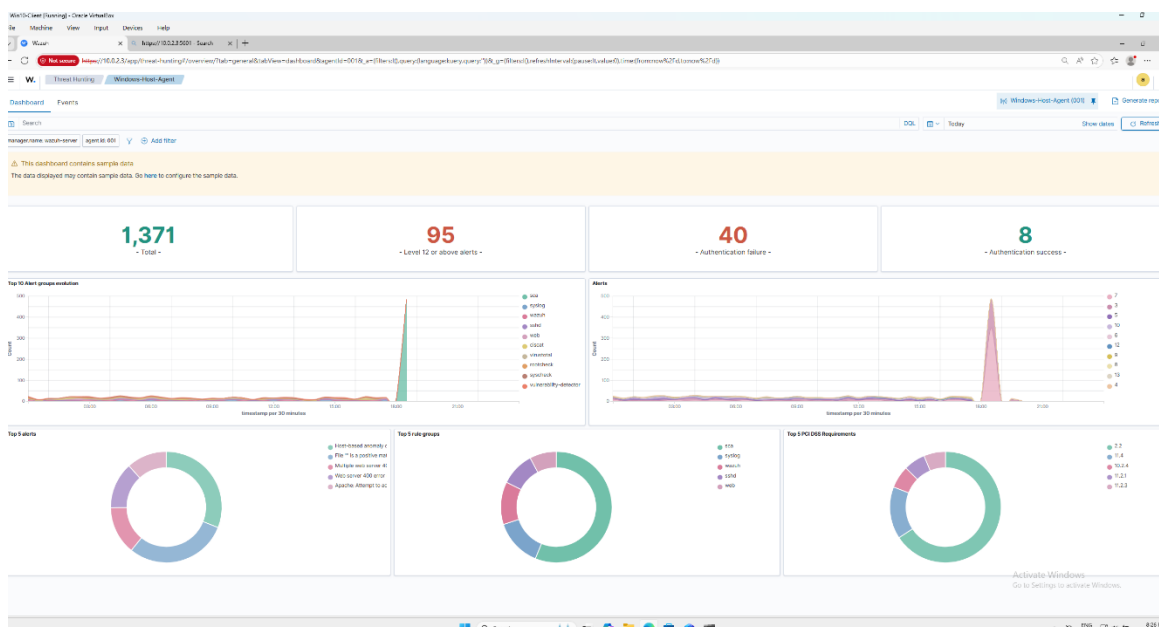
1. On your Windows host, open Command Prompt or PowerShell
2. Attempt to SSH into the Wazuh VM with incorrect credentials multiple times:


```
ssh wazuh-user@10.0.2.3
```
3. When prompted for password, enter an incorrect password deliberately
4. Repeat this 5-10 times with wrong passwords
5. This repeated failed authentication attempt should trigger an alert on the Wazuh server for brute-force attack detection

Step 12: View Generated Alerts in Dashboard

1. In the Wazuh Dashboard, click the menu icon (three horizontal lines)
2. Navigate to **Threat Hunting** section (or **Security Events** depending on Wazuh version)
3. The Threat Hunting page displays recent security events and alerts

4. Set the time filter to show recent events:
 - Click the time range selector at the top
 - Select **Last 15 minutes** or **Last 30 minutes**
5. Look for alerts related to:
 - FIM (File Integrity Monitoring) alerts with rule name containing "File"
 - Failed login attempts if you used Method 2
6. Click on any alert to view detailed information:
 - Alert timestamp
 - Source IP address (usually localhost or the agent IP)
 - Rule description
 - Alert level (severity)
 - Full log details
7. The alerts should clearly indicate:
 - **Alert Time:** When the event occurred
 - **Agent Name:** Windows-Host-Agent
 - **Rule Description:** Details about what triggered the alert
 - **Level:** Severity level (typically 3-7 for file modifications)
 - **Event Data:** Complete details of what was monitored



Threat Hunting Dashboard - Generated Alerts

Lab 2: Hybrid Cloud Deployment (Wazuh-as-a-Service)

Objective

To transition from local on-premise management to a cloud-based SaaS (Software-as-a-Service) model, integrating the local asset monitored in Lab 1 with Wazuh Cloud, and performing security compliance assessments using the cloud infrastructure.

Section 1: Cloud Account Setup

Step 1: Register for Wazuh Cloud Free Trial

1. Open a web browser and navigate to:

<https://console.cloud.wazuh.com/sign-up>

2. You will see the Wazuh Cloud registration form. Fill in the following information:

- **Email Address:** Enter a valid email address (use the same email if you have a Wazuh account)
- **Password:** Create a strong password (minimum 8 characters, including uppercase, lowercase, numbers, and special characters)
- **First Name:** Your first name
- **Last Name:** Your last name
- **Company:** Enter your university name or "Student Lab"
- **Country:** Select your country
- **Terms of Service:** Check the box to accept the terms

3. Click the **Start free trial** button

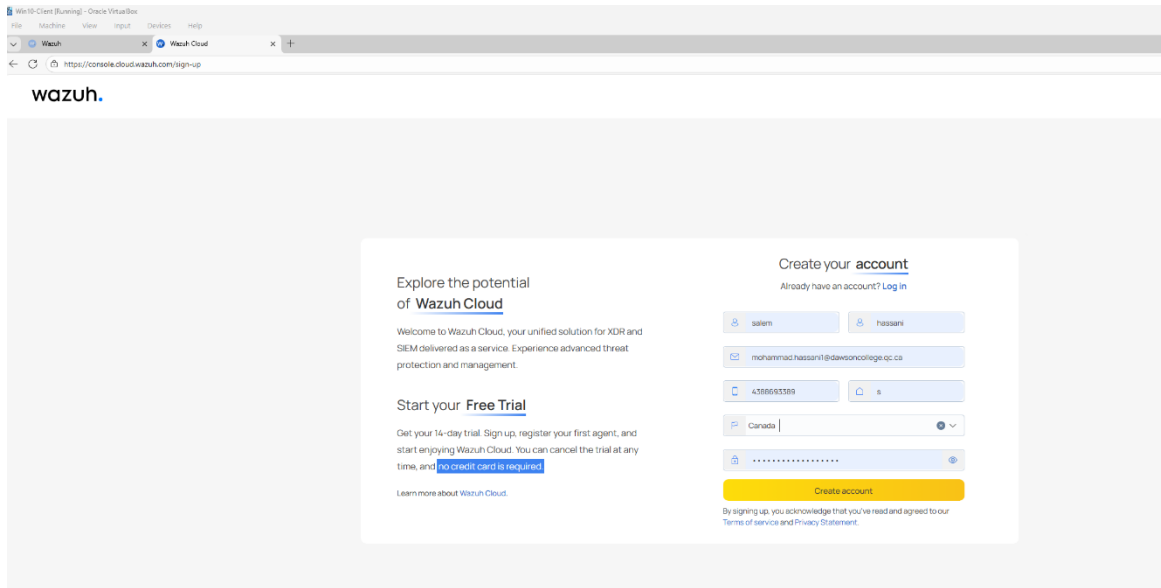
4. An email will be sent to your registered email address

5. Open your email and find the verification link from Wazuh

6. Click the verification link in the email (it may ask you to verify your email or log in)

7. After verification, you will be redirected to the Wazuh Cloud Console

Important Note: No credit card is required for the free trial. The trial is valid for 14 days and provides full access to cloud features.



Wazuh Cloud Registration Page

Step 2: Create Cloud Environment

1. Log in to the Wazuh Cloud Console at:

<https://console.cloud.wazuh.com/>

(You should be automatically logged in after email verification)

2. On the main console page, locate and click the **Create environment** button
3. The environment creation wizard will appear. Configure the following settings:
 - **Environment Name:** Lab-Test-Environment (or any descriptive name)
 - **Use Case:** Select **Education/Learning** or **Testing/Development**
 - **Region:** Select the region closest to your geographic location for optimal latency
 - **Environment Profile:** Select **Small** (suitable for laboratory use and up to 100 active agents)
4. Review the Small Profile specifications:
 - Maximum 100 active agents
 - 1 month of indexed data retention
 - 3 months of archived data retention
 - 100-500 events per second capacity
5. Leave **Pricing** on **Monthly** (display only - your trial account is free)

6. Review the cost estimate (it will show as free for trial accounts)
7. Click **Start your free trial** or **Create environment** button
8. The console will display a progress indicator showing the environment is being provisioned
9. Wait for the environment to be fully created and initialized (this typically takes 5-10 minutes)
10. The environment status will change from "Creating" or "Provisioning" to **Ready**
11. Once Ready, you will see the environment dashboard with:
 - Environment name
 - Environment URL (endpoint for agents to connect to)
 - Status: Ready
 - Creation date and time

The image shows two screenshots of the Wazuh Cloud Console. The top screenshot is the 'Create your environment' page, and the bottom screenshot is the 'Environment details' page for a created environment named 'Lab-Test-Environment'.

Create your environment

Name your new environment: Lab-Test-Environment

Region: Canada Central (Canada Central)

What is your use case for the trial?

Select your profile: Small (selected), Medium, Large, Custom

Basic settings: [Reset](#)

- Active agents: 100
- Indexed data retention: 1 month
- Archive data retention: 3 months
- Support plan: Standard support

Advanced settings (Optional):

- Average / Peak EPS: 100-500 (Use recommended value)
- Indexed data capacity: 20 GB (Use recommended value)

Settings review:

- The following settings are limited during the trial period: up to 100 active agents, 500 Avg/Peak EPS and 20 GB of indexed data retention.
- This environment can accommodate up to 100 active agents, enabling the processing of over 4.5 million events each day. You will be able to explore and analyze security alerts for up to 3 months, with an archival period of 3 months. [Read More](#)

Select your pricing:

Free trial enabled

Free 14-day trial. No credit card required.

MONTHLY: \$623/month (Billed monthly, Annual price: \$7476)

ANNUAL: \$5711/year (Billed annually, Save 6% over the monthly price)

[Start your free trial](#)

Environment details

Lab-Test-Environment

Status: Ready

Lab-Test-Environment | Edit name | Canada Central (Canada Central) | ewnu7258nc | 2026/01/05

Settings | All Reports | Metrics

⚠ This is a trial environment. During the trial period some settings are limited. Purchase environment to remove the limits.

- Active agents: Up to 100
- Indexed data retention: 1 month
- Average / Peak EPS: 100-500
- Archive data retention: 3 months
- Support plan: Standard
- Indexed data capacity: 20 GB

[Back to environments list](#)

Wazuh Cloud Console - Environment Ready

Section 2: Network and Connectivity Verification

Step 3: Verify Internet Connectivity and Required Ports

Before deploying agents to the cloud, verify that your system can reach the cloud endpoints:

1. On your Windows host machine, open PowerShell (regular user mode is fine for testing)
2. Test connectivity to the Wazuh Cloud endpoint by running:

`Test-NetConnection -ComputerName console.cloud.wazuh.com -Port 443`

3. The output should show:

`TcpTestSucceeded : True`

4. This confirms you can reach the cloud service on port 443 (HTTPS)
5. Verify the three critical ports needed for agent communication are accessible:
 - **Port 1514 (TCP):** Used for agent-to-manager communication
 - **Port 1515 (TCP):** Used for agent enrollment/registration
 - **Port 443 (TCP/HTTPS):** Used for web dashboard and API communication
6. Test each port (if your network firewall allows, some test port attempts may be blocked):

`Test-NetConnection -ComputerName console.cloud.wazuh.com -Port 1514`

`Test-NetConnection -ComputerName console.cloud.wazuh.com -Port 1515`

`Test-NetConnection -ComputerName console.cloud.wazuh.com -Port 443`

7. At minimum, port 443 should show "TcpTestSucceeded : True"
8. If all tests succeed, your network connectivity is properly configured for cloud agent deployment

```
PS C:\Users\Administrator> Test-NetConnection -ComputerName console.cloud.wazuh.com -Port 443

ComputerName      : console.cloud.wazuh.com
RemoteAddress     : 13.225.196.76
RemotePort        : 443
InterfaceAlias    : Ethernet
SourceAddress     : 10.0.2.4
TcpTestSucceeded  : True
```

Network and Connectivity Verification443

Section 3: Cloud Agent Deployment

Step 4: Prepare Local Agent for Cloud Migration

If you still have the Wazuh Agent installed from Lab 1 on your Windows host:

Option A: Stop the Local Agent Service (Recommended)

1. Open PowerShell as Administrator
2. Stop the Wazuh Agent service:

NET STOP WazuhSvc

3. The service will stop, preventing the local agent from reporting
4. This keeps the agent installed but inactive, allowing easy reactivation if needed

Option B: Completely Uninstall the Local Agent

1. Open **Control Panel > Programs and Features**
2. Find and select **Wazuh Agent** in the installed programs list
3. Click **Uninstall**
4. Follow the uninstall wizard and confirm removal
5. Once uninstalled, the agent is completely removed from the system

I did the first options(A)

A screenshot of a PowerShell terminal window. The prompt is 'PS C:\Users\Administrator>'. The command entered is 'NET STOP WazuhSvc'. The output is 'The Wazuh service was stopped successfully.' The prompt is now 'PS C:\Users\Administrator> |'.

```
PS C:\Users\Administrator> NET STOP WazuhSvc
The Wazuh service was stopped successfully.
PS C:\Users\Administrator> |
```

NET STOP WazuhSvc

Step 5: Deploy New Agent to Cloud Wazuh

I wanted to inform you that my Wazuh Cloud account, which I created using my college email, was unexpectedly suspended after one day. Since using non-business emails (such as personal ones) requires additional identity verification, I am unable to continue with the lab for now after this step.

Thanks

Your trial is over



Your environment is **terminated**.
Create a new environment to continue using Wazuh Cloud.

You will be able to start a new trial on **2026-04-20**.

Create new environment

Terminated environments

Name	Status	Region	Creation date	Termination date
Lab-Test-Environment ID: eiwnu17268mo	Terminated	Canada Central (Canada Central)	2026/01/05	2026/01/06